

Xi Zhong

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Expected Ph.D. Completion: May 2027

Research Interests

My research lies at the intersection of information and coding theory, distributed computing, and large-scale computing systems. I combine fundamental analysis with algorithm and system design to develop efficient, reliable, and resource-aware computing methods, and evaluate them in practical computing environments.

Education

Ph.D. in Electrical and Computer Engineering

Aug. 2021 – May 2027 (expected)

University of Florida, Gainesville, FL, USA

Aug. 2024 – present

University of Utah, Salt Lake City, UT, USA

Aug. 2021 – Aug. 2024

Advisor: Prof. Mingyue Ji

Moved with my advisor's research group from the University of Utah to the University of Florida in Aug. 2024; Ph.D. degree to be conferred by the University of Florida.

M.Eng. in Computer Science and Technology

Sep. 2018 – Jun. 2021

Guangxi Normal University, Guilin, China

Advisor: Prof. Minquan Cheng

B.S. in Statistics

Sep. 2014 – Jun. 2018

Guangxi Normal University, Guilin, China

Publications

Conference Papers

- [C1] **X. Zhong**, J. Kliewer, and M. Ji, “Feasibility of Coded Polynomial Aggregation under Straggler Patterns,” *IEEE Information Theory Workshop (ITW)*, Tempe, AZ, USA, Nov. 2026, accepted (to appear).
- [C2] **X. Zhong**, J. Kliewer, and M. Ji, “Fundamental Limits of Coded Polynomial Aggregation,” *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, China, Jun. 2026.
- [C3] **X. Zhong**, S. Lu, J. Kliewer, and M. Ji, “Uncoded Download in Lagrange-Coded Elastic Computing with Straggler Tolerance,” *IEEE International Symposium on Information Theory (ISIT)*, 2025.
- [C4] **X. Zhong**, S. Lu, J. Kliewer, and M. Ji, “Dual-Lagrange Encoding for Storage and Download in Elastic Computing for Resilience,” *IEEE International Symposium on Information Theory (ISIT)*, 2025.
- [C5] **X. Zhong**, J. Kliewer, and M. Ji, “Uncoded Storage Coded Transmission Elastic Computing with Straggler Tolerance in Heterogeneous Systems,” *IEEE International Conference on Communications (ICC)*, 2024.
- [C6] **X. Zhong**, J. Kliewer, and M. Ji, “Matrix Multiplication with Straggler Tolerance in Coded Elastic Computing via Lagrange Code,” *IEEE International Conference on Communications (ICC)*, 2023.

Journal Articles

- [J1] M. Cheng, J. Wang, **X. Zhong**, and Q. Wang, “A Framework of Constructing Placement Delivery Arrays for Centralized Coded Caching,” *IEEE Transactions on Information Theory*, vol. 67, no. 11, pp. 7121–7131, Nov. 2021.

- [J2] **X. Zhong**, M. Cheng, and R. Wei, “Coded Caching Schemes with Linear Subpacketizations,” *IEEE Transactions on Communications*, vol. 69, no. 6, pp. 3628–3637, Jun. 2021.
- [J3] **X. Zhong**, M. Cheng, and J. Jiang, “Placement Delivery Array Based on Concatenating Construction,” *IEEE Communications Letters*, vol. 24, no. 6, pp. 1216–1220, Jun. 2020.
- [J4] M. Cheng, Y. Li, **X. Zhong**, and R. Wei, “Improved Constructions of Coded Caching Schemes for Combination Networks,” *IEEE Transactions on Communications*, vol. 68, no. 10, pp. 5965–5975, Oct. 2020.
- [J5] M. Zhang, M. Cheng, J. Wang, **X. Zhong**, and Y. Chen, “Improving Placement Delivery Array Coded Caching Schemes with Coded Placement,” *IEEE Access*, vol. 8, pp. 217456–217462, 2020.

Manuscripts

- [M1] **X. Zhong**, J. Kliever, and M. Ji, “Feasibility of Coded Polynomial Aggregation,” manuscript.
- [M2] **X. Zhong**, J. Kliever, and M. Ji, “Uncoded Storage and Coded Download for Heterogeneous Elastic Matrix Multiplication,” manuscript.
- [M3] **X. Zhong**, J. Kliever, and M. Ji, “Optimal Computation Assignment in Lagrange Coded Download Elastic Computing for Heterogeneous Systems,” manuscript.

Selected Research Contributions

Distributed Aggregation and Function Computation

Developed coded methods for directly computing aggregate functions in distributed systems under stragglers, avoiding unnecessary recovery of individual intermediate results. Established fundamental feasibility and recovery limits through necessary and sufficient conditions and explicit constructions [C1, C2].

Elastic and Heterogeneous Cloud Computing

Designed computation-assignment schemes for distributed systems with heterogeneous machine speeds, storage capacities, elasticity, and stragglers. Established theoretical performance guarantees and implemented the schemes on AWS EC2, reducing completion time by up to 30% relative to prior coded assignments and by over 50% relative to uncoded baselines under stragglers [C3–C6].

Efficient Content Delivery and Caching

Developed coded caching constructions with substantially reduced subpacketization, addressing a key implementation-complexity bottleneck in coded content delivery [J1–J5].

Presentations

Talks

- [T1] *Feasibility of Coded Polynomial Aggregation under Straggler Patterns*, IEEE Information Theory Workshop (ITW), Tempe, AZ, USA, Nov. 2026 (upcoming).
- [T2] *Uncoded Storage Coded Transmission Elastic Computing with Straggler Tolerance in Heterogeneous Systems*, IEEE International Conference on Communications (ICC), Denver, CO, USA, Jun. 2024.

Posters and Demonstrations

- [P1] *Fundamental Limits of Coded Polynomial Aggregation* (poster), North American School of Information Theory (NASIT), Brigham Young University, Provo, UT, USA, Jun. 2026.
- [P2] *Uncoded Storage Coded Transmission Elastic Computing with Straggler Tolerance in Heterogeneous Systems* (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2024.
- [P3] *Lagrange-Coded Elastic Computing on AWS EC2* (live demonstration), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2024.

Teaching and Mentoring

Research Mentoring

Samuel Lu, High School Student Researcher, Rowland Hall, Salt Lake City 2024 – 2025
Mentored an independent research project on straggler-tolerant coded elastic computing, covering Lagrange coding fundamentals, scheme design, and numerical evaluation. The work resulted in co-authorship on two IEEE ISIT 2025 papers [C3, C4].

Teaching

Teaching Assistant Fall 2022, Fall 2023
ECE 3500: Fundamentals of Signals and Systems
University of Utah
Instructor: Prof. Mingyue Ji
Graded assignments and examinations and held office hours to assist students with course material.

Honors and Awards

IEEE Information Theory Workshop (ITW) Travel Grant	2026
Outstanding Graduate, Guangxi Normal University	2021

Professional Service

Technical Reviewer

Journals:
IEEE Transactions on Information Theory (TIT)
IEEE Transactions on Communications (TCOM)
EURASIP Journal on Wireless Communications and Networking (JWCN)
Conferences:
IEEE International Symposium on Information Theory (ISIT)
IEEE International Conference on Communications (ICC)
IEEE Information Theory Workshop (ITW)

Volunteer

Student Volunteer, Warren B. Nelms Annual IoT Conference, University of Florida	2024, 2025
Student Volunteer, IEEE International Conference on Computer Communications (INFOCOM)	2022

Technical Skills

Programming and Computing: Python, MATLAB, distributed computing experimentation on AWS EC2
Languages: Mandarin Chinese (native); English (fluent)